

Noun technique analysis

Step 1: Identify All Nouns from Use Cases, Actors, and System Information

- Actors: Doctor, Patient, Nurse, Insurance Representative, Pharmacist
- The nouns in the system vision: Patient records, Appointment, claims, billing, Medication orders, Inventory levels, Prescriptions, Treatment plans, health Status, Medical history, Insurance coverage, Vital signs

Inputs and outputs:

- Inputs: Claims, Medication orders, Patient updates
- Outputs: Patient information, Appointment reminders, Treatment status, Medical history, Prescriptions, Billing statements, Inventory reports

Noun	Source
Doctor	Actor
Patient	Actor
Nurse	Actor
Insurance Representative	Actor
Pharmacist	Actor
Patient Record	System Vision
Appointment	System Vision
Claim	System Vision
Billing	System Vision
Medication Order	System Vision
Inventory	System Vision
Prescription	System Vision
Treatment Plan	System Vision
Health Status	System Vision
Medical History	System Vision
Insurance Coverage	System Vision
Vital Sign	System Vision
Reminder	Output
Statement	Output
Report	Output

Step 2: Add Additional Items Based on Existing Systems

Additional nouns that may be relevant for the healthcare management system:

Noun	Justification/Source
Diagnosis	Essential for patient recordkeeping
Dosage	Needed in prescriptions
Payment Method	Relevant for claims and billing
Address	Relevant to Patient Information
Contact Information	Important for patient management

Step 3: Refining the list by evaluating each noun

refined list:

Noun	Include/Exclude/Research	Reason
Doctor	Include	Unique role with relevant permissions
Patient	Include	Central to system, requires multiple records
Nurse	Include	Has access to patient vitals and updates
Insurance Representative	Include	Essential for claims processing
Pharmacist	Include	Manages prescriptions and inventory
Patient Record	Include	Core function
Appointment	Include	Critical for scheduling
Claim	Include	Needed for insurance processing
Billing	Include	Necessary for financial transactions
Medication Order	Include	Essential for fulfilling prescriptions
Inventory	Include	Helps manage stock
Prescription	Include	Key to patient treatment
Treatment Plan	Include	Helps in consistent care
Health Status	Include	Monitored by healthcare team

Medical History	Include	Important for informed treatment
Insurance Coverage	Include	Necessary for claims
Vital Sign	Include	Regularly updated by nurse
Reminder	Include	Supports patient engagement
Statement	Include	Important for billing and claims
Report	Include	Relevant for management
Diagnosis	Include	Crucial for treatment plans
Dosage	Include	Specific detail in prescriptions
Payment Method	Research further	Could be part of the billing or claims process
Address	Research further	Might be part of patient information
Contact Information	Research further	Part of the patient record

Step 4: Master list with inclusion status

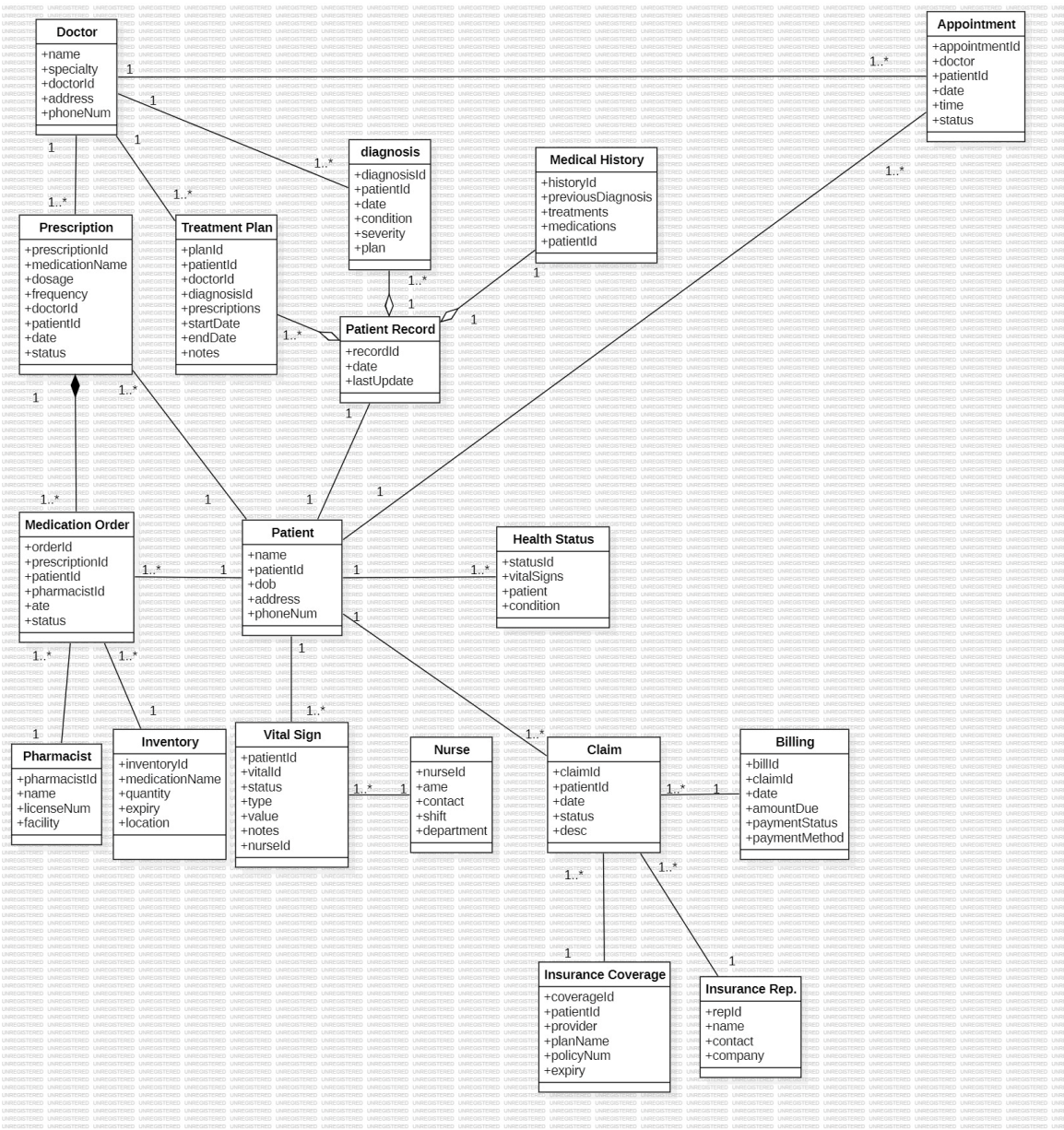
Master list based on the analysis:

Noun	Inclusion status
Doctor	Include
Patient	Include
Nurse	Include
Insurance Representative	Include
Pharmacist	Include
Patient Record	Include
Appointment	Include
Claim	Include
Billing	Include
Medication Order	Include
Inventory	Include
Prescription	Include
Treatment Plan	Include
Health Status	Include
Medical History	Include
Insurance Coverage	Include
Vital Sign	Include
Reminder	Include
Statement	Include
Report	Include
Diagnosis	Include
Dosage	Research further
Payment Method	Research further

Address	Research further
Contact Information	Research further

Last step we should review with users, stakeholders and team members to define the list of things in the problem domain.

Domain Model Class Diagram:



Crud technique for use cases

1. Book appointment:

- C: Patient schedules a new appointment with a doctor.
- R: Patient or doctor views appointment details.
- U: Patient reschedules or modifies the appointment. If wanted
- D: Patient cancels the appointment.

2. Prescribe medication:

- C: Doctor adds a new prescription for a patient.
- R: Patient views prescribed medications in the patient's record.
- U: Doctor adjusts dosage or medication details.
- D: Doctor removes a prescription if it's completed.

3. Process claim:

- C: insurance representative adds a new claim for a patient.
- R: Representative or patient reviews claim details.
- U: Representative updates billing information.
- D: Representative removes a claim if necessary.

4. Fulfill medication order:

- C: Pharmacist records a new medication order
- R: Pharmacist reviews the medication details and prescription.
- U: Pharmacist updates order status ex. "ready for pickup"
- D: Pharmacist cancels an order if it's invalid or no longer needed.

5. View medical records:

- R: Doctor, nurse, or patient view their medical history, treatments, and prescriptions.

6. View patient vitals:

- C: Nurse records new vitals in the patient's chart.
- R: Nurse views and monitors a patient's vitals and care notes.
- U: Nurse updates vitals if new readings occur.
- D: Nurse removes incorrect vitals entries or healed entries if needed.

7. Review patient claims:

- C: Insurance representative adds comments on a claim.
- R: Representative reviews details of submitted patient claims.
- U: Representative approves or denies claim status
- D: Representative removes a claim if it's invalid.

8. Update inventory levels:

- C: Pharmacist records newly received inventory in the system.

- R: Pharmacist reviews current medication inventory levels.
- U: Pharmacist adjusts inventory after dispensing medications.
- D: Pharmacists remove records of expired medications.

9. Update medical records:

- C: Doctor adds new medical information to a record.
- R: Doctor views existing details in the patient's medical record.
- U: Doctor modifies or updates information in the patient's record.
- D: Doctor removes outdated or incorrect information as necessary.

10. Update patient information:

- C: Patient submits personal information if setting up an account.
- R: Patient views current personal information on file.
- U: Patient updates details such as address or phone number.
- D: Patient can remove outdated information

11. Generate appointment report:

- R: Management views a report showing total appointments.
- D :Management clears or removes past reports if needed.

12. Generate claims report:

- R: Management views a report of all claims processed
- D: Management removes past reports if no longer needed.

13. Generate inventory report:

- R: Management views the current stock levels for medications.
- D: Management removes the report after review.

14. Generate patient report:

- R: Management views how many patients a doctor has
- D: Management deletes the report if needed

15. Generate payroll statements:

- R: System generates and management views payroll statements.
- D: Management archives the payroll statements after review.

16. Send monthly billing statements:

- R: System retrieves the billing details for each patient to include in the statement and send it to them

17. Send appointment reminders:

- R: System retrieves appointment details for patients scheduled the next day

18. Send treatment status report:

- R: System retrieves treatment status and patient details.

19. Update appointment status:

- R: System retrieves the current status of the appointment.
- U: System updates the status of the appointment as in canceled or confirmed

20. Update prescription status:

- R: System retrieves the current prescription status.
- U: System updates the prescription status to indicate it has been refilled.

21. Update patient health status:

- R: System retrieves the current health status
- U: System updates the patient's health record with new vitals or changes.

Updated Use Case diagram:

